Andrew Desmedt

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EXPERIENCE

STEL Controls - Fredericksburg, VA — *Junior Mechatronics Engineer*

August 2025 – August 2026

- Collaborated with the engineering team on Python and C++ control software for a precision mechatronic system achieving sub-millimeter repeatability, contributing to motion planning and external subsystem communication.
- Supported the system's integration with a FANUC CRX-10iA/L collaborative robotic arm, Rockwell PLC, and AI-based vision.
- Extensively developed custom end-effectors for the robotic arm, including TCP (tool center point) setup and calibration.
- Conducted prototyping, testing, and troubleshooting with linear, air, and piezo actuators.
- Documented code, processes, and system configurations.
- Debugged embedded hardware and communication interfaces (I2C, USB, serial) using an oscilloscope during bring-up and troubleshooting.
- Integrated laser triangulation distance sensors for high-precision position feedback, supporting the system's sub-millimeter repeatability.
- Designed and prototyped a 3-nozzle paint-jet head and custom three-port manifold.
- Performed CAM programming and CNC machining with 3D toolpaths.
- Programmed microcontrollers, developed circuits with over-the-air updates, and gained early hands-on exposure to PCB design.
- Wired PoE Security cameras with RJ45 crimped ends.

Team 1731 Fresta Valley Robotics — Captain, Mentor

Captain of robotics team - 2025; Mentor - 2026

- Led a competitive robotics team, coordinating mechanical, electrical, and software subsystems.
- Wired and configured CAN bus networks for motor controllers and sensors across the robot's electrical subsystem.
- Helped test QuestNav, one of the first teams to trial Meta Quest VR headset-based robot localization.
- Won four district events, a district championship, and competed in world championship two years in a row.
- Mentored younger team members.
- Came up with a solution that enabled the robot to complete an extremely valuable task.

SKILLS

Java, Python, C++

Microcontrollers: ESP32, STM32 Arduino

CAD/CAM - Onshape, Fusion, Inventor, AutoCAD

GitHub

CNC Machining

I^2C, Ethernet, CAN, USB, Serial

Microsoft Office

Troubleshooting

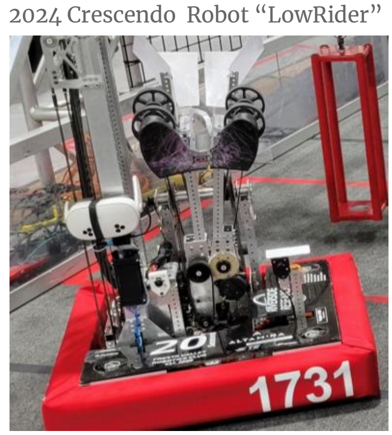
3d Printing: FDM/Resin

Customer service

Sales & Marketing

Project Management

2024 Crescendo Robot "LowRider"



2025 Reefscape Robot "MacDyver"

PROJECTS

GPS-Guided Drone — *Personal Project*

In Progress

- Build a custom FPV-style quadcopter from individually sourced parts: motors, flight controller, ESC, ELRS radio receiver, and a GPS module.
- Hand-solder all power and signal wiring between the flight controller, ESC, and motors.
- Run INAV flight control firmware with GPS enabled for position hold and waypoint-capable navigation.
- Showcased the build at the Laurel Ridge Community College STEM Fair.

ADDITIONAL EXPERIENCE

Andy's Heritage Hens - Nokesville, VA — *Owner/Manager*

April 2019 - May 2023

- Managed care of up to 70 laying hens.
- Delivered fresh eggs to 20 customers weekly.
- Managed all aspects of the business, including budgeting, marketing, customer service, and logistics every day for 4 years.
- Built a custom chicken coop with an automatic coop door to improve efficiency. This was my first experience with robotics.


EDUCATION

Virginia Polytechnic Institute and State University — *Bachelor of Applied Science in Electrical Engineering*

July 2026 – Present

- Future coursework includes PCB design with Altium Designer.

Laurel Ridge Community College, Warrenton — *Associate of Applied Science in Engineering*

September 2023 - May 2026

- Designed, wired, and programmed a remote-control vehicle.
- Coursework includes mechanical systems, electronics, and CAD.
- Graduated Summa Cum Laude.

CERTIFICATES

ETA International Fiber Optic Installer

GPS-Guided Drone

AWARDS

2025 Innovation in control Award First Chesapeake Glenn Allen VA Event

2025 Winner First Chesapeake Glenn Allen VA Event

2025 Winner First Chesapeake Bethesda MD Event

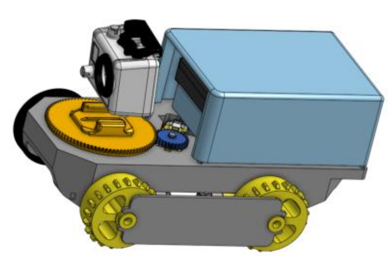
2026 Awardee Outstanding Achievement in Engineering and Engineering Technology, Laurel Ridge Community College

LANGUAGES

Mandarin HSK-4 eligible
Russian A1

HONOR SOCIETIES

Epsilon Pi Tau
Phi Theta Kappa



Laurel Ridge Design Project